

Week 5 — Effect modification and interaction: learn the distinction, then stress-test it

Duration: ~2.5 h (Tue 6 Oct, before the Thu 8 Oct lab). Deck: `lectures/slides/Week5_interaction_slides.qmd`. Its MASLD numbers come from `lectures/analysis/Week5_joint_exposure.R` (written to `Week5_joint_exposure.json`) and from the P3 and M1 result objects (`examples/_results/`). The code and output on its “In R” slides come from `lectures/analysis/Week5_code_outputs.R`, which runs each block exactly as shown and records what R printed.

The design. Teach the clean 2×2 idea first, on a made-up fixed-follow-up example (smoking, income, hypertension), where the two questions can be read straight off the table. Then use MASLD as the **stress test**: the same two questions on the paper’s four phenotype groups, where time order, sparse cells and selection make the causal reading hard. R code sits next to the concept it computes; every causal assumption, target population and contrast is stated on the slide.

Timing. The deck’s speaker notes give **142 minutes** of content and no break slot. With the usual 10-minute break (placed below after Slide 18, before “Count first”) that is **152 minutes — 2 over the 150-minute class**. Take the two minutes from *If you are running behind* at the foot of this file. If you retime a slide, retime it in the deck’s notes too; this plan copies their minutes.

Built on Week 4. Same rows (**5,911**), same pre-exposure adjustment set L (age, sex, race/ethnicity, smoking, sedentary time), same DAG style, Week 4’s collapsibility table reread for modification (Slide 13), and Week 4’s g-computation idea extended to standardised survival (Slides 23–26).

Row-set discipline. Every quantitative slide names its rows. - **5,911** — Week 4’s rows: the locked domain complete on every covariate of the paper’s Model 2 and the pre-exposure set. All of the MASLD analysis in Part 3 and Part 4’s interaction slides. - **6,048** — the locked domain itself: only the crude IV-vs-I table by sex (Slides 33–35), the rows P3’s code uses. - **1,296** — P3’s rows (Groups III + IV complete on Model 1) and **1,291** (Groups III + IV within the 5,911): the sex test on Slide 36.

What this week does NOT do. Survey weights are Week 6: every number here is **unweighted**. RERI, AP and S are in the appendix and the lab, not the core lecture. Propensity-score methods for subgroup effects (Karim, ch. 8, §8.6) come after the propensity-score weeks.

Learning objectives

By the end, students answer the deck’s five questions, in order (Slide 2):

1. Am I changing **one exposure** (effect modification) or imagining **two exposures jointly** (interaction)?
2. What is my modifier M, and could **A change M**?
3. On what **scale** am I comparing effects — additive or multiplicative?
4. What do the **four cells** actually contain?
5. What should I **report**, with uncertainty, without over-interpreting it?

...and can write the R for each step: cell counts, stratum-specific HRs from a product term and the likelihood-ratio test, standardised six-year risks, and the interaction contrast.

Continuity

- **Last week (Week 4 / L3).** Confounding with obesity as the only exposure; the DAG decides what to adjust; ORs and HRs move without confounding (collapsibility).
- **Today.** The paper’s real exposure is obesity $\times$ waist fat. Two questions on one 2×2 .

- **Thursday (L4).** EpiMethods `confoundingE2`: the RHC data, a **binary** outcome, DNR as the modifier, logistic regression, RERI/AP/S. See *Thursday* below.
 - **P3.** One pre-specified modifier on the student's own paper (optional, ungraded; enters M1). Slides 10 and 38 are the P3 checklist.
-

Opening · 6 minutes

Slide 1 — Two questions hiding in the same 2×2 table (4 min)

- The four groups with people and deaths: **I 355 / 4 · II 4,265 / 362 · III 732 / 67 · IV 559 / 101** (5,911 rows). Question 1, effect modification: does waist fat's effect differ by obesity? Question 2, interaction: what do the two together do?
- Point at Group I's 4 deaths, but do not unpack the causal problems yet.

Slide 2 — The map for today (2 min)

- The five questions above; the lecture's recurring algorithm. "A model term such as $A * M$ comes after these questions."

Part 1 · Learn the distinction on a clean example · 24 minutes

Slide 3 — Start with smoking, income, and hypertension (3 min)

- A fixed-follow-up example: A = smoking, M = low income **measured before** the smoking window, Y = incident hypertension. Risks 5% / 15% / 10% / 30%. Say the time order out loud: it is what makes income a defensible modifier here.

Slide 4 — Question 1: effect modification (4 min)

- One exposure, smoking; income defines the groups. RD +5 vs +15 points (modified on the additive scale); RR 2.0 in both (not modified on the multiplicative scale). Hence **effect-measure modification**.

Slide 5 — In R: read smoking's effect within each income group (3 min)

- A data frame and two lines: RD and RR by income. The code does exactly what the table did.

Slide 6 — Question 2: interaction (5 min)

- Two exposures, reference = higher income, not smoking. RR 2, 3, 6 = 2×3 : no multiplicative interaction. **IC** defined at first use as a difference of risk differences: $(30 - 15) - (10 - 5) = +10$ **points**; null IC = 0.
- The causal reading is stronger: intervening on income is less well-defined than using baseline income to define strata.

Slide 7 — In R: interaction on two scales (3 min)

- `mult_ratio` = 1 and IC = 0.1 (ten percentage points). RERI/AP/S are deliberately not introduced here (appendix and lab).

Slide 8 — Same table, different question (3 min)

- The anchor slide: effect modification imagines changing one variable; interaction, two. Point back to it whenever MASLD gets complicated.

Slide 9 — Confounding is a third idea (3 min)

- Control confounding; describe modification; describe all four exposure states for interaction. A product term is a tool, on its model's scale; the question comes first.

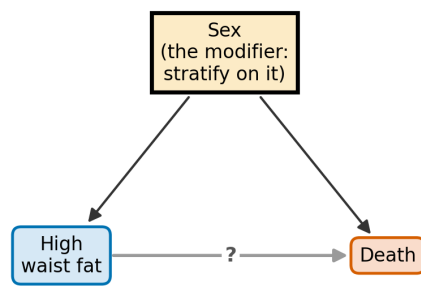
Part 2 · A practical workflow for effect modification · 21 minutes

Slide 10 — A five-step workflow (3 min)

- Choose A and M with a reason; check A cannot cause M; count $A \times M$ cells; estimate A within each M on a named scale; report both effects, their contrast and uncertainty, and the adjustment set. **This is the P3 workflow.**

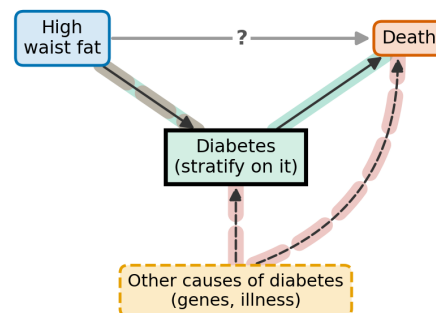
Slide 11 — The modifier must survive the “could A change it?” test (5 min) · do not drop

A · Fine: the exposure cannot change it



Waist fat cannot change anyone's sex: each stratum holds the same people, whatever their waist size.

B · Not fine: the exposure can change it



Stratifying on diabetes (1) blocks part of waist fat's effect (green) and (2) opens waist fat → diabetes ← other causes → death (red).

Square = stratified on. Dashed outline = not measured. Grey ? = the effect we want. Green = part of the effect; red = an open non-causal path.

- “A’s effect among people with $M = m$ ” is cleanest when A cannot move people into or out of M. Sex, age, genes, or a condition established before A are natural candidates; a condition measured at the same visit may not be.
- If A could cause M, stratifying may block part of A’s effect (green in the figure) or induce collider bias (red). **Timing is the practical check, not the definition:** the rule is that M must not be affected by A.

Slide 12 — What do we adjust for? (5 min)

- Effect modification: control confounding of $A \rightarrow Y$ within levels of M; the causal effect of M itself need not be identified, but conditioning on M must be causally defensible.
- Interaction: the **joint** effects of A and M must be identified — a stronger requirement. The adjustment set follows the question, not the presence of $A * M$ in software.

Slide 13 — Name the scale before saying “no modification” (4 min)

- Week 4’s collapsibility table read for modification: OR 4.00 in both age groups, RD 0.30 in both, RR 1.60 vs 2.50. Never say “no effect modification” without naming the scale.

Slide 14 — Two scales are enough for the core lecture (4 min)

- Multiplicative: ratio of HRs (null 1). Additive: difference of RDs (null 0). For survival data, use standardised risks at a fixed time for the additive scale.

Part 3 · Now stress-test the idea with MASLD · 53 minutes, with the break after Slide 18

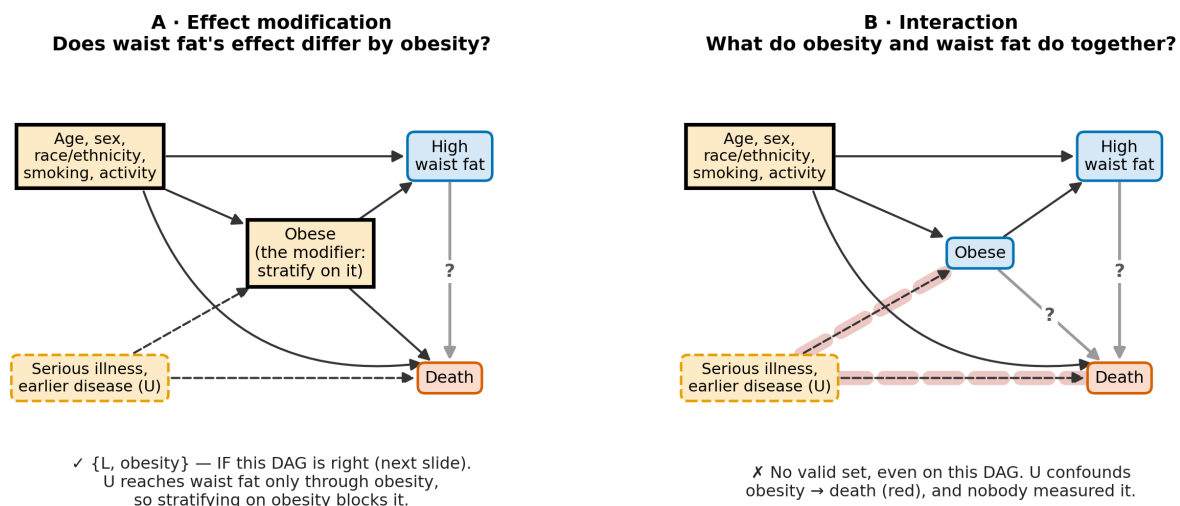
Slide 15 — The MASLD question we will try to answer (2 min)

- A = high waist fat ($\text{WHtR} \geq 0.6$); M = obesity; Y = all-cause death; L = Week 4's pre-exposure set. Primary question: does waist fat's effect on death differ between people with and without obesity? The paper asked a **prognostic phenotype** question; the causal re-reading is ours.

Slide 16 — Why the MASLD example becomes causally difficult (4 min)

- BMI and waist were measured at the **same examination**; both are adiposity measures, so “obesity before waist fat” is an assumption; MASLD ascertainment depends partly on BMI and waist; Group I is sparse — the paper reports **5 deaths** (386 people; its Table 2), our complete-case rows 4. Nothing here makes the paper wrong.

Slide 17 — Causal audit: can obesity really be the modifier? (7 min) · do not drop



Square = conditioned on (adjusted, stratified, restricted or selected). Dashed outline = not measured. Grey arrow with ? = the effect we want.

- Panel A (effect modification): {L, obesity} is the minimal set **on this DAG**, under strong assumptions, including obesity before waist fat. Panel B (interaction): U confounds obesity → death and is unmeasured, so the joint effect has no valid set even then.
- “The question changed, so the identification problem changed.” dagitty-checked; the deck's appendix prints the DAG as dagitty code.

Slide 18 — What the effect-modification interpretation assumes (5 min)

- Obesity established before the waist-fat exposure; no unmeasured common cause of obesity and waist fat that stratifying on obesity would open; L adequate within each stratum; conditioning on MASLD not inducing important selection bias. FLI includes BMI **and** waist circumference (plus TG and GGT),

so the last cannot be ruled out. Say it is an identification concern we cannot rule out, not that the bias is large.

BREAK (10 min) — not in the deck's notes; put it here

Slide 19 — Count first (3 min)

- Group I anchors the obese waist-fat comparison II vs I and has **4 deaths**. Among 2,432 obese women only **57** have lower waist fat. Report the cell counts beside the estimates.

Slide 20 — In R: count before you model (3 min)

- `table(group, died)` and the obese-women table: two lines students reuse in P3.

Slide 21 — One product-term model, two stratum-specific HRs (4 min)

- `fit <- coxph(... ~ obese * central + L)`. Not obese: **1.09 (0.79–1.50)**; obese: **2.99 (1.11–8.09)**; ratio **2.74 (0.97–7.71)**; likelihood-ratio $p = 0.03$. The product term re-parameterises the same four cells; it does not create information where Group I has four deaths.

Slide 22 — In R: recover both HRs and test the product term (4 min)

- `exp(b["central"]) = 1.09` and `exp(b["central"] + b["obese:central"]) = 2.99`; `anova(fit_main, fit)` gives the likelihood-ratio p , 0.029.

Slide 23 — Why compute standardised six-year risks? (4 min)

- Our re-analysis, not the paper's. Same people, waist fat set lower then high, predicted six-year risk averaged — Week 4's g-computation with a survival model.

Slide 24 — Additive scale: the standardised six-year results (3 min)

- Obese 2.3% $\rightarrow$ 6.4% (**+4.1 points**); not obese 9.1% $\rightarrow$ 9.8% (**+0.7**). Difference of risk differences **+3.4 (–0.1 to +6.6)** points (500 bootstrap resamples, 10 of which drew no Group I deaths). Model-standardised risks, not observed proportions.

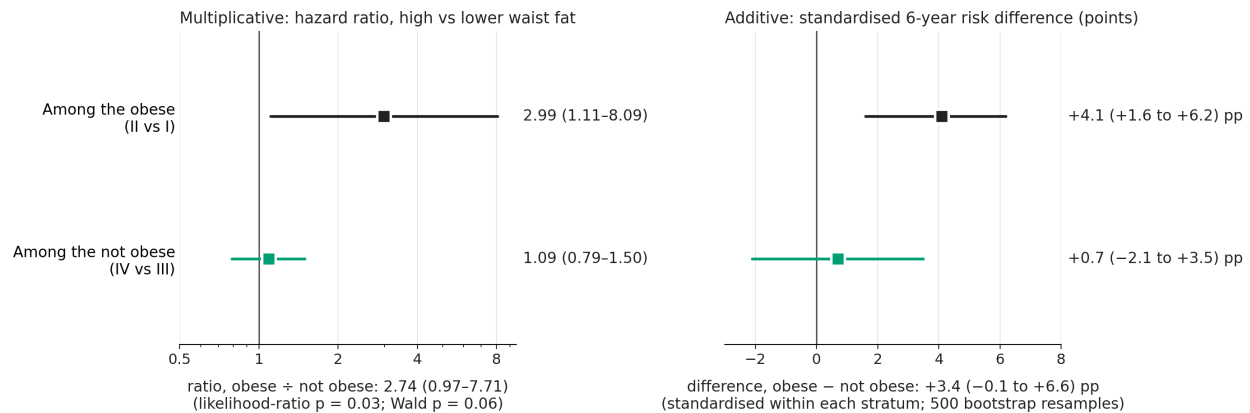
Slide 25 — In R: use the same people twice, changing only waist fat (4 min)

- `risk6()` copies a stratum, changes only `central`, predicts survival at six years and averages: 2.3, 6.4, 9.1, 9.8%.

Slide 26 — In R: turn the four predictions into risk differences (2 min)

- 4.1 and 0.7 points; difference 3.4.

Slide 27 — Put both scales on one page (4 min)



- The report: waist fat **may** have a larger effect among the obese on both scales, but the estimate is unstable and rests on a reference cell with four deaths. Avoid “significant interaction” as a headline.

Slide 28 — The paper’s IV vs I comparison asks a different question (4 min)

- II vs I and IV vs III change waist fat only (within-row); IV vs I changes both (diagonal): useful for phenotype risk stratification, not the effect of high waist fat among the not obese. Do not decompose it through alternative routes in the core lecture.

Part 4 · Interaction, survival tools, and reporting · 38 minutes

Slide 29 — If obesity is a second exposure, the estimand changes (5 min)

- Keep the coding: A = high waist fat, M = obesity; four potential outcomes $Y(a, m)$. Multiplicative: **2.74 (0.97–7.71)**. Additive: with all four joint states standardised to one common population, **IC = +3.9 points**. A causal joint-exposure reading also needs obesity’s effect identified, and U is unmeasured.

Slide 30 — In R: interaction with the same A and M coding (2 min)

- The multiplicative contrast is the same product term: $\exp(b["obese:central"]) = 2.74$.

Slide 31 — In R: the additive interaction contrast (2 min)

- `risk6_all()` sets everyone to each joint state: R00 6.9, R10 7.5, R01 2.5, R11 7.0% (III, IV, I, II); IC = 3.9 points. Why 3.9 here and 3.4 before: effect modification standardises within each obesity stratum; joint interaction standardises all four states to one common population.

Slide 32 — The right survival tool depends on the scale (4 min)

- Multiplicative: Cox $A * M \rightarrow$ ratio of HRs, **2.74 (0.97–7.71)**. Additive: standardised risk at a fixed time $\rightarrow$ difference of RDs, **+3.4 (–0.1 to +6.6) points**. Do not turn a censored outcome into “ever died”; a fixed-follow-up binary outcome (Thursday) is different.

Slide 33 — A dramatic subgroup claim can be built on almost no information (3 min)

- Crude IV vs I by sex, 6,048 rows: men **36.14 (8.86–147.48)**, women **4.11 (0.99–17.12)**; Group I deaths / people **2 / 317** and **2 / 58**. Diagonal phenotype comparisons, not sex-specific waist-fat effects. No biological story.

Slide 34 — In R: reproduce the crude IV-vs-I HRs (3 min)

- `relevel(..., ref = "I")`, one crude Cox model per sex; `groupIV` is IV vs I.

Slide 35 — In R: now count the reference events (2 min)

- Group I by sex: 2 deaths in men, 2 in women.

Slide 36 — Ask the subgroup question cleanly (4 min)

- The locked IV-vs-III contrast with sex as the modifier: crude **0.62 (0.26–1.43)**, p 0.278 (P3); the paper’s Model 1 **1.00 (0.42–2.35)**, p 0.993 (P3); Week 4’s pre-exposure set **1.13 (0.48–2.66)**, p 0.78. Six-year RDs about +0.8 points in men and +0.5 in women. “No evidence” is not “evidence of no difference”.

Slide 37 — In R: one test for sex modification on the locked contrast (3 min)

- Restrict to III + IV; `IV * female` with L; ratio 1.13 (0.48–2.66), likelihood-ratio p 0.78.

Slide 38 — Honest subgroup reporting (3 min)

- Pre-specify and say why A cannot cause M; name the question; count every cell; A within each M on a named scale; contrast and interval, not just p; no invented biology.

Slide 39 — The algorithm to take into your own paper (3 min)

- One exposure or two? Can A change M? Which scale? What do the cells contain?

Slide 40 — Key takeaways (4 min)

- Six takeaways; exit question: *one modifier for your own paper, and one sentence on why your exposure cannot change it*. **Next:** Week 6, survey weights — design-aware, P3’s Model 1 sex ratio becomes **2.53 (0.92–6.98)** (M1).

Appendix (uncounted)

- One model, two parameterisations (`group obese * central, all.equal TRUE`).
- RERI, AP, S and IC with their nulls; RERI/AP/S depend on the reference coding and are not emphasised because obesity’s joint causal effect is not identified here.
- Why diabetes is a risky modifier: diabetes in 13% vs 28% of people with lower vs high waist fat. What splitting by diabetes produces: waist fat’s HR **1.23** without diabetes and **0.93** with it — neither a clean causal subgroup effect if waist fat can affect diabetes.
- Where to read more; references; the two questions as dagitty code.

Thursday — L4 (EpiMethods confoundingE2) · for the TA and the instructor, not in the deck

The deck carries no lab bridge; the L4 handout carries the short version of these notes. They are worth five minutes at the start of Thursday’s session.

- **Data.** The RHC data again (Week 4’s L3): the 1,439 complete cases, restricted to `DASI > 20` → **623** patients. Outcome: the binary death variable, **40%** died. Exposure: `no_rhc_status`; modifier / second exposure: `no_dnr_status`.

- **The exercise’s reference cell.** The exercise text says it changes the reference levels “to No RHC and No”; its code does the reverse. Both indicators are coded 1 for “No”, so the reference cell (0, 0) is **RHC and a DNR order** — the lowest-risk cell, which RERI needs (Knol et al. 2011).
- **Count first — that reference cell holds 3 patients.** Only **23** of the 623 have a DNR order (20 without RHC, 3 with). Run `with(high_dasi_data, table(rhc_status, dnr_status, death_status))`: the RHC + DNR cell has 3 patients and 1 death. Everything in Problem 2 divides by it:
 - OR for no RHC among DNR patients **29.8 (1.14–781)**;
 - RERI **–28.5 (–125 to 68)**, AP **–14.0**, S **0.04 (0.002–0.80)**;
 - product term $p = 0.056$ (Problem 1’s $p = 0.048$, with RHC’s confounders only).
 - The only well-supported stratum estimate is no RHC among patients **without** a DNR order: OR **1.21 (0.78–1.86)**, 600 patients. This is Tuesday’s “Count first” in the lab’s data: Group I has 4 deaths; the lab’s reference cell has 3 patients.
- **Read the intervals, not interactionR’s p column.** `interactionR` (0.1.7) computes the RERI, AP and S p-values one-sided or on the wrong scale: S prints $p = 0.98$ while its own interval (0.002–0.80) excludes 1. The exercise tells students to compare those p-values with 0.05.
- **interactionR prints the wrong way round in Problem 1.** With `em = TRUE` it treats the **first** name in `exposure_names` as the modifier, so as called it shows DNR’s effect by RHC — the reverse of Problem 1’s question. Read Problem 1 from the Publish rows (no RHC vs RHC: **33.0 (1.79–609)** with a DNR order, **1.22 (0.80–1.85)** without), or swap the order of `exposure_names`.
- **Statements in the exercise text to correct out loud:**
 1. “SI = 1: no multiplicative interaction” — S is an **additive** measure (deck appendix).
 2. “RERI < 0 ... the exposures may be protective when combined” — a negative RERI means **less than additive**, not protective.
 3. Interaction as “a departure from multiplicativity” — differs from the *Concepts* page and from Tuesday (the joint effect of two exposures, on either scale).
- **Death is common (40%),** so ORs overstate RRs and the OR-based RERI is not the risk-based RERI.
- **The 1,439 are complete cases,** which Week 4 showed is selection on early survival (the ADL score). It carries into L4 unchanged.
- **Is DNR a fair modifier?** DNR and RHC are both day-1 decisions, and which came first is not recorded. Tuesday’s rule: A must not be able to change M. If catheterisation (or what it revealed) could prompt a DNR order, DNR fails it, and Problem 1’s reading assumes it could not.
- To reproduce: run the solution’s two `glm()` calls on `rhc_data.rds` from `confoundingEx2.zip`, plus `interactionR::interactionR()` and `Publish::publish()` as the solution calls them.

If you are running behind

The notes run 142 minutes; with the break, 152. Cut in this order.

1. **Slides 30–31 — interaction in R (4 → 1 min).** Say “same product term; IC 3.9 over one common population” and move on.
2. **Slides 34–35 — the crude IV-vs-I code (5 → 2 min).** Show the counts slide only.
3. **Slide 23 — why standardise (4 → 2 min).** Slide 25’s code walks through it anyway.
4. **Slide 9 — confounding is a third idea (3 → 1 min).**

Never drop Slides 11, 17, 19 or 27. The could-A-change-M rule, the causal audit, the cell counts and the two-scale summary are the argument.

Reading map — each segment and where students can review it

Segment	EpiMethods / course page	Key reference
Definitions (Slides 3–9)	<code>confounding0</code> <i>Concepts</i> , “Formal Definitions”; effect-modification video and slides	VanderWeele 2009; Bours 2021; Karim ch. 8 §8.2–8.5
The modifier rule, adjustment sets (Slides 11–12, 17–18)	<code>confounding0</code> , “Implications for Confounding Control”	Karim ch. 8, Fig. 8.1
Scale (Slides 13–14)	<code>confounding0</code> , “The Role of the Scale”; <code>confounding5</code> (Week 4)	VanderWeele & Knol 2014
Joint variable vs product term; RERI, AP, S (appendix)	<code>confounding9</code> ; interaction tutorial	Knol et al. 2011; Hosmer & Lemeshow 1992
Reporting (Slides 38–39)	<code>confounding0</code> , “Reporting guideline”	Knol & VanderWeele 2012; Karim ch. 8 §8.7
Thursday	<code>confoundingE2</code> , <code>confoundingE2solution</code>	—

Before assigning the reading.

- **Karim, ch. 8.** Sections 8.1–8.5 and 8.7 match today; §8.6 belongs after the propensity-score weeks. Three slips in the assigned sections: §8.4.3.1 gives the ratio of odds ratios as 1.27 “with a 95% confidence interval of 1.04 to 1.27” (Table 8.3: 1.04–1.56); the same paragraph calls $OR_{M=0} = 2.93$ “the low-income group” although §8.3 codes high income as $M = 0$; and §8.5.1 describes AP as the share “attributed to the exposure” rather than to the interaction.
- **The interaction tutorial** states that RERI “ranges from 0 to ∞ ” and AP “from -1 to 1 ”. Both can be negative, and AP can fall below -1 .

Appendix — the two questions as dagitty code

```
dag {
  Waist [exposure]  Death [outcome]  U [latent]
  L -> Obese  L -> Waist  L -> Death
  U -> Obese  U -> Death
  Obese -> Waist  Obese -> Death  Waist -> Death
}
```

With **Waist** as the only exposure dagitty returns $\{L, Obese\}$. Add **Obese** as a second exposure and no set exists; remove the latent tag from U and $\{L, U\}$ appears. Each of Slide 18’s assumptions, broken, removes Panel A’s set: reverse **Obese** $\rightarrow$ **Waist**; add **U** $\rightarrow$ **Waist**; add a latent **G** $\rightarrow$ **Obese**, **G** $\rightarrow$ **Waist**; add a latent income node; or add **FLI** [adjusted] **Waist** $\rightarrow$ **FLI** **Waist** $\rightarrow$ **TG** **TG** $\rightarrow$ **FLI** **TG** $\rightarrow$ **Death**.

References

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- Knol MJ, VanderWeele TJ, Groenwold RHH, et al. Estimating measures of interaction on an additive scale for preventive exposures. *Eur J Epidemiol* 2011;26:433–438.
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- VanderWeele TJ. On the distinction between interaction and effect modification. *Epidemiology* 2009;20:863–871.
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